

Document 07: Database Design & Business Rules Specification

Project: Natural Intellects School Management System (NISMS)

Version: 1.0

Status: Draft

Purpose:

This document defines the database architecture, entity relationships, constraints, business rules, auditing principles, and data lifecycle management strategies for the Natural Intellects School Management System.

This document serves as the primary reference for:

- Prisma schema generation
- PostgreSQL implementation
- Cloud AI development
- Backend services
- Database migrations
- System testing

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1. DATABASE DESIGN PRINCIPLES

Historical Preservation

The system shall avoid hard deletion.

Preferred approaches:

- Status changes
 - Soft archiving
 - Immutable transactions
-

Normalization

The database shall minimize duplication while maintaining acceptable performance.

Target:

Third Normal Form (3NF).

Strong Relationships

Foreign keys shall enforce:

- Student relationships
 - Teacher relationships
 - Academic relationships
 - Financial relationships
-

Auditability

Critical actions shall remain traceable.

Examples:

- Payment creation
- Student transfers
- Promotions

- Academic approvals

2. NAMING CONVENTIONS

Table Names

Recommended:

Plural Pascal Case in Prisma:

```
model Student
```

```
model Teacher
```

```
model Payment
```

Database Tables

Generated:

```
students
```

```
teachers
```

```
payments
```

Primary Keys

Standard:

```
id UUID PRIMARY KEY
```

Timestamps

Every major table shall include:

createdAt

updatedAt

Archive Fields

Recommended:

status

archivedAt

archiveReason

3. CORE ENTITIES

SCHOOL

Purpose:

Represents the school installation.

Fields:

id

name

shortName

email

phone

address

logoUrl

websiteDomain

createdAt

updatedAt

Business Rules:

One deployment shall have:

ONE SCHOOL RECORD.

USER

Purpose:

Authentication and permissions.

Fields:

id

email

passwordHash

role

isActive

lastLoginAt

createdAt

updatedAt

Roles:

SUPER_ADMIN

PROPRIETOR

HEAD_TEACHER

SECRETARY

BURSAR

TEACHER

STUDENT

Purpose:

Stores student information.

Fields:

id

admissionNumber

firstName

middleName

lastName

gender

dateOfBirth

nationality

religion

photoUrl

admissionDate

status

classId

streamId

createdAt

updatedAt

Statuses:

ACTIVE

TRANSFERRED

GRADUATED

ARCHIVED

Business Rules:

- Admission numbers must be unique.
 - Students cannot be deleted.
 - Students belong to one class.
 - Students belong to one stream.
-

GUARDIAN

Purpose:

Stores parent and guardian information.

Fields:

id

fullName

phoneNumber

alternativePhone

relationship

address

occupation

createdAt

updatedAt

Relationships:

One Guardian

↓

Many Students

Support:

- Father
 - Mother
 - Sponsor
 - Relative
-

STUDENT GUARDIAN

Purpose:

Many-to-many relationship.

Fields:

studentId

guardianId

isPrimaryContact

Business Rules:

Students may have multiple guardians.

TEACHER

Purpose:

Stores staff information.

Fields:

id

staffNumber

firstName

middleName

lastName

gender

dateOfBirth

photoUrl

phoneNumber

email

address

employmentDate

employmentType

status

createdAt

updatedAt

Employment Types:

FULL_TIME

PART_TIME

CONTRACT

Statuses:

ACTIVE

ON_LEAVE

ENDED

ARCHIVED

Business Rules:

Teachers cannot be permanently deleted.

CLASS

Purpose:

Academic levels.

Examples:

P1

P2

S1

S2

Fields:

id

name

level

createdAt

updatedAt

STREAM

Purpose:

Class streams.

Examples:

P4 Blue

P4 Red

S2 East

Fields:

id

classId

name

createdAt

updatedAt

Business Rules:

A stream belongs to one class.

SUBJECT

Purpose:

Academic subjects.

Fields:

id
name
code
category
createdAt
updatedAt

Categories:

SCIENCE
ART
LANGUAGE
OTHER

TEACHER SUBJECT

Purpose:

Many-to-many relationship.

Fields:

teacherId
subjectId

TEACHER CLASS

Purpose:

Subject teaching assignments.

Fields:

teacherId

classId

subjectId

isClassTeacher

Business Rules:

One class shall have:

ONE CLASS TEACHER.

4. ACADEMIC ENTITIES

TERM

Purpose:

Academic terms.

Fields:

id

name

year

startDate

endDate

isActive

Examples:

TERM_I
TERM_II
TERM_III

ATTENDANCE

Purpose:

Daily student attendance.

Fields:

id
studentId
teacherId
date
status
remarks
createdAt

Statuses:

PRESENT
ABSENT
SICK
EXCUSED

Business Rules:

One student:

↓

One attendance record per day.

EXAM

Purpose:

Assessment records.

Fields:

```
id
name
termId
classId
totalMarks
createdAt
```

Examples:

```
Mid-Term
End-Term
Mock Exams
```

STUDENT MARK

Purpose:

Academic performance.

Fields:

id

studentId

subjectId

examId

teacherId

score

grade

remarks

Business Rules:

Scores:

- Cannot be negative.
- Cannot exceed total marks.

REPORT CARD

Purpose:

Generated reports.

Fields:

id

studentId

termId

generatedAt

generatedBy

pdfUr1

Business Rules:

Finalized reports become immutable.

5. FINANCIAL ENTITIES

FEE STRUCTURE

Purpose:

School fee definitions.

Fields:

```
id
classId
termId
amount
description
```

PAYMENT

Purpose:

Student payments.

Fields:

```
id
```

studentId
amount
paymentMethod
referenceNumber
receiptNumber
paymentDate
recordedBy
createdAt

Payment Methods:

CASH
MOBILE_MONEY
BANK

Business Rules:

Payments:

- Cannot be deleted.
- Cannot be edited after confirmation.

Corrections:

↓

Adjustment records.

PAYMENT ADJUSTMENT

Purpose:

Financial corrections.

Fields:

```
id
paymentId
reason
amountDifference
createdBy
createdAt
```

Business Rules:

Every adjustment requires:

- User information
- Reason
- Timestamp

6. WEBSITE ENTITIES

NEWS ARTICLE

Purpose:

Website news management.

Fields:

```
id
title
slug
content
featuredImage
```

publishedAt

createdBy

GALLERY IMAGE

Purpose:

Photo galleries.

Fields:

id

title

imageUrl

description

uploadedAt

PAGE CONTENT

Purpose:

Editable pages.

Examples:

ABOUT

ADMISSIONS

ACADEMICS

Fields:

id

pageType

content

updatedAt

7. RELATIONSHIP RULES

School

↓

Students

Teachers

Classes

Subjects

Class

↓

Many Streams

Student

↓

Many Attendance Records

Student

↓

Many Payments

Student

↓

Many Guardians

Teacher

↓

Many Subjects

Teacher

↓

Many Classes

8. DATA INTEGRITY RULES

Student Rules

Students:

- Cannot be deleted.
 - Must belong to one class.
 - Must belong to one stream.
-

Teacher Rules

Teachers:

- Cannot be deleted.

- May teach multiple subjects.
-

Financial Rules

Payments:

- Immutable after confirmation.
-

Academic Rules

Finalized report cards:

- Cannot be modified.
-

Attendance Rules

One record:

↓

Per student

↓

Per day

9. ARCHIVING STRATEGY

Students

Use:

status

archivedAt

archiveReason

Teachers

Use:

```
status  
archivedAt
```

Financial Records

Never archived.

Always retained.

Academic Records

Never deleted.

Always retained.

10. AUDIT REQUIREMENTS

Future Audit Table

Recommended:

```
AUDIT_LOG
```

Fields:

```
id  
userId  
action  
entityType
```


entityId

oldValues

newValues

timestamp

Examples:

Created Student

Transferred Student

Recorded Payment

Generated Reports

11. FUTURE DATABASE EXPANSION

Potential future entities:

Parent Portal

Examples:

PARENT_ACCOUNT

SMS Integrations

Examples:

SMS_LOG

Library Systems

Examples:

BOOK

BORROW_RECORD

Inventory Systems

Examples:

INVENTORY_ITEM

Hostel Systems

Examples:

HOSTEL

DORMITORY

BED_ASSIGNMENT

STRATEGIC CONCLUSIONS

The NISMS database shall prioritize:

- Strong relational integrity
- Historical preservation
- Simplicity
- Auditability
- Future scalability

The database design shall support:

- Cloud AI implementation
- Prisma schema generation
- PostgreSQL best practices
- Long-term maintainability

This document serves as the authoritative source for all future database development activities.

End of Document 07.